A High-Performance Rule-Based Multi-Agent Framework for Spatio-Temporal Vector Dynamics: Application to *Anopheles stephensi* in Ethiopia

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Abstract

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1 Introduction

Vector-borne diseases (VBDs) represent one of the most profound challenges to global health security, accounting for approximately 700,000 deaths annually and imposing significant economic burdens on developing nations (Lowe and Codeço, 2025). Malaria alone threatens nearly half the world's population, with Sub-Saharan Africa bearing 94% of the global case burden as (Organization, 2024). While the past two decades witnessed substantial reductions in malaria mortality through coordinated interventions, this hard-won progress faces unprecedented threats from ecological and demographic shifts. The rapid urbanization of Sub-Saharan Africa, currently the world's fastest urbanizing region (Sulemana et al., 2019), has created novel transmission ecologies that challenge historical assumptions about vector ecology and control strategies (Joseph et al., 2024).

The primary driver of this epidemiological transition is the globalization of previously localized vectors, particularly the invasive malaria mosquito *Anopheles stephensi*. Unlike endemic African malaria vectors that predominantly breed in natural water bodies, *An. stephensi* exhibits remarkable ecological plasticity, thriving in artificial containers and man-made structures characteristic of urban environments (Hunter et al., 2025). This species shares critical adaptive traits with *Aedes aegypti*, the primary dengue vector, including container breeding, anthropophily, and exophilic resting behaviors (Hamlet et al., 2022). The consequences of this adaptation were discored in Djibouti, where *An. stephensi* establishment triggered a 30-fold increase in malaria cases within seven years, transforming a nation on the verge of malaria elimination into a high-transmission zone (Hamlet et al., 2022). This vector's capacity to exploit previously low-risk urban settings necessitates fundamentally new approaches to surveillance, modeling, and intervention design (Mwema et al., 2024).

A critical bottleneck in responding to this emerging threat is the pervasive "Data Gap" characterizing many resource-constrained settings where *An. stephensi* is establishing. Effective vector control relies on robust decision-support systems built upon high-quality, high-resolution entomological and epidemiological data, yet national surveillance programs often operate with limited financial and technological resources (Lowe and Codeço, 2025). Current modeling approaches struggle to bridge the gap between theoretical elegance and practical utility: empirical models propagate measurement errors through unreliable data streams, while process-based models frequently lack the spatial granularity and computational efficiency needed for operational deployment (Lowe and Codeço, 2025). This disconnect is particularly acute for invasive vectors, where absence of historical baseline data undermines traditional statistical approaches (Pascoe et al., 2019).

To address these challenges, we propose a novel computational framework that integrates high-performance computing methodologies with ecological process modeling. Our system implements a layered, rule-based Multi-Agent Simulation (MAS) architecture that harmonizes heterogeneous spatio-temporal data streams into a unified analytical environment. This framework is part of the family of ABS (Agent Based Simulation) which are simulation models where the researcher explicitly describe the decision processes of simulated actors at micro-level (Siebers and Aickelin, 2008). The proposed framework bridges Computer Science and Environmental Science by conceptualizing environmental data as dynamic information layers, including static rasters (for instance: elevation, building density, population density), time-series climate variables, and vector data (shapefiles), that

interact with autonomous agent populations. We distinguish between two agent classes: "Inert Agents" representing static environmental features (urban water containers serving as breeding sites) and "Living Agents" modeling biological vectors (An. stephensi mosquitoes for example) with behavioral rules derived from empirical studies.

This study demonstrates how such synthetic modeling approaches can provide critical insights where field data are scarce or unreliable. Using Dire Dawa, and Somali woredas from Ethiopia, which are regions experiencing *An. stephensi* invasion, (Zhou et al., 2024; Hawaria et al., 2023) as a proof-of-concept, we simulate 6 months of mosquito population dynamics across the whole regions. Our results illustrate how agent-based modeling can identify high-risk micro-habitats, simulate seasonal population trajectories, and potentially assist intervention strategies in data-limited settings. By making the simulation framework open-source and computationally efficient, we aim to empower public health agencies in resource-constrained regions with decision-support tools capable of addressing the unique challenges posed by urban-adapted invasive vectors.

2 Materials and Methods

2.1 Framework Architecture

2.1.1 Overview

The simulation framework follows a modular, agent-based design that integrates spatially explicit environmental data, temperature-dependent life-cycle models, and behavioural rules. It is implemented in Java and built around four core subsystems: (i) a time manager that advances simulation time in discrete ticks (15-minutes per tick) and interpolates climate data; (ii) a spatial registry that indexes agents on a lazy grid for efficient neighbourhood queries; (iii) an agent layer system that manages agents and evaluates rules in parallel; and (iv) a lifecycle model that computes temperature-driven development and survival rates. The overall architecture is illustrated in Figure 1.

```
  <!-- Mermaid chart: framework architecture -->
graph TD
    A[TimeManager] --> B[SimulationEngine]
    B --> C[SpatialRegistry]
    B --> D[AgentLayer: Mosquitoes]
    B --> E[AgentLayer: WaterTanks]
    D --> F[RuleEngine]
    E --> F
    D --> G[LifecycleModel]
    B --> H[Raster Layers (Climate, Static)]
    B --> I[Export Service]
    I --> J[CSV Snapshots]
    J --> K[Analysis Notebook]
```

Figure 1: High-level architecture of the simulation framework.

2.1.2 Core Components

Agent layers. Two agent types are implemented: `LivingAgent` (representing a mosquitoe) and `InertAgent` (representing a breeding site). Living agents track life stage (egg, larva, pupa, adult), age, energy, gravid status, and resting behaviour. Their movement is modelled as a random walk with a step size derived from field observations of flight range (Sayeed Quraishi et al., 1966). Inert agents maintain water volume, egg count, larval count, and carrying capacity; they update their capacity dynamically based on water level, mimicking the density-dependent mortality observed in larval habitats (Thomas et al., 2016). Agent layers are processed in parallel using a partition-based container (`AgentContainer`) and a rule-engine that evaluates behavioural rules (e.g., feeding, oviposition, resting) in priority order (Ogunlade et al., 2023).

Raster layers and climate data. Static raster layers (elevation, building density, population density) are loaded from GeoTIFF files using GeoTools and stored either in memory or memory-mapped for large grids (Balkew et al., 2021). Climate time series (2-m temperature and total precipitation) are ingested from NetCDF files (ERA5-Land reanalysis) into interpolated raster layers that perform linear temporal interpolation between hourly steps (Chabot-Couture et al., 2014; Merkord et al., 2017).

Lifecycle model. The lifecycle model represents mosquito population dynamics using a discrete-time, stage-structured Metzler matrix (Ogunlade et al., 2023; Agboka et al., 2025b). Four developmental stages are tracked: eggs (E), larvae (L), pupae (P), and adults (A). All rates are temperature-dependent and defined on a per-day basis, then converted to tick-level probabilities via the Poisson process ($p = 1 - \exp(-r\Delta t)$, where Δt is the tick duration in days). The daily projection matrix is:

$$\mathbf{M}(T) = \begin{pmatrix} (1-d_E)S_E & 0 & 0 & F \cdot K \\ d_E S_E & (1-d_L)S_L & 0 & 0 \\ 0 & d_L S_L & (1-d_P)S_P & 0 \\ 0 & 0 & d_P S_P & 1-\mu_A \end{pmatrix}$$

with stage abundances evolving as $[E, L, P, A]_{t+1}^\top = \mathbf{M}(T)[E, L, P, A]_t^\top$. The temperature-dependent functions are:

- **Development rates** (day^{-1}): $d_X(T) = \max(0, a_X T^2 + b_X T + c_X)$ for $X \in \{E, L, P\}$.
- **Stage survival** (Gaussian): $S_X(T) = \text{amp}_X \cdot \exp\left(-\frac{1}{2}\left(\frac{T-\mu_X}{\sigma_X}\right)^2\right)$.
- **Adult mortality** (day^{-1}): $\mu_A(T) = \max(0, a_A T^2 + b_A T + c_A)$.
- **Fecundity** (eggs/female/day): $F(T) = \max\left(0, a_F e^{b_F T} - e^{b_F T_{\max} - \left(\frac{T_{\max}-T}{c_F}\right)^2}\right)$.
- **Host carrying capacity**: $K = 1 + \ln(1 + \text{population density})$, which scales fecundity based on human population density.
- **Density-dependent larval mortality**: each aquatic habitat has a carrying capacity cap (function of water volume); if larval count exceeds cap, excess larvae are removed probabilistically.

All temperature-response coefficients are stored in a YAML configuration file under the `species` key, making the model generic and easily adaptable to any mosquito species by replacing the coefficients in Table 1. This parametrisation approach follows the principles of generic insect population modelling (Agboka et al., 2025a,b, 2026).

Table 1: Generic temperature-response parameters for a mosquito lifecycle model.

Parameter set	Meaning	Unit	YAML key
d_E coeff.	Egg development rate	$1/°C^2, 1/°C, 1$	`egg_dev_a,b,c`
d_L coeff.	Larva development rate	$1/°C^2, 1/°C, 1$	`larva_dev_a,b,c`
d_P coeff.	Pupa development rate	$1/°C^2, 1/°C, 1$	`pupa_dev_a,b,c`
S_E	Egg survival (amp, mean, sigma)	dim., °C,° C	`egg_survival_...`
S_L	Larva survival (amp, mean, sigma)	dim., °C,° C	`larva_survival_...`
S_P	Pupa survival (amp, mean, sigma)	dim., °C,° C	`pupa_survival_...`
μ_A coeff.	Adult mortality	$1/°C^2, 1/°C, 1$	`adult_mort_a,b,c`
F coeff.	Fecundity	$a_F, b_F, T_{\max}, c_F$	`fecundity_a,b,Tmax,c`
K	Host scaling factor	dimensionless	(derived)

Rule engine and spatial registry. Behavioural rules (e.g., "`feed`", "`lay_eggs`", "`rest_in_building`", etc.) are defined in a YAML configuration file and evaluated using a JavaScript engine (GraalVM). The spatial registry implements a lazy grid with a configurable cell size (default $0.001/°C \approx 110m$). Neighbour queries are performed by checking cells within a Chebyshev distance $\lceil \text{radius}/\text{cellSize} \rceil$ and cache results for one second to reduce computational cost.

2.1.3 Simulation Configuration

All simulation parameters are defined in a single YAML file (`simulation.yaml`). The file specifies:

- **Time settings**: start date, total number of ticks, tick duration (15-minutes).
- **File paths**: study area shapefile, elevation, building density, population rasters, and the climate NetCDF.
- **Species parameters**: coefficients for development, survival, fecundity, and adult mortality.

- **Tokens**: mapping between environment variable names (e.g., "temperature") and raster layers.
- **Agent layers and rules**: each layer contains a list of rules with a condition, an action, and a priority.
- **Seeding**: number of breeding sites and mosquitoes, habitat grid resolution, and whether to seed uniformly or using a weighted habitat suitability map.
- **Project defaults**: agent search radius, step size, maximum age.

At startup, the `App` class loads the configuration, initialises the time manager, loads rasters and climate data, creates the spatial registry, builds agent layers, seeds the initial population, and starts the simulation engine.

2.1.4 Result Analysis

Simulation outputs are exported as CSV snapshots every 100 ticks (25 days). Each snapshot contains per-agent records with tick count, agent ID, layer, type, coordinates, alive flag, age, stage, energy, gravid status, and for inert agents, egg/larval counts and water volume. Environmental values (temperature, precipitation, etc) are attached to each agent's row via bilinear interpolation from the current climate frame.

The analysis notebook (Python) performs the following steps:

1. **Daily aggregation**: The CSV is read in chunks (100 000 rows) to compute daily adult, larva, and pupa counts, as well as tank statistics (mean water volume, total eggs/larvae), using the tick-to-day conversion (96 ticks per day).
2. **Population dynamics**: Stage counts are plotted, and stage durations are estimated from cross-correlation of smoothed population peaks. Theoretical durations are computed from species parameters at the mean observed temperature.
3. **Environmental correlation**: Pearson correlation between daily adult counts and mean temperature / precipitation is calculated.
4. **Suitability mapping**: A kernel density estimate (KDE) of adult points (bandwidth 0.002° 220 m) is generated and exported as a GeoTIFF. A breeding site potential raster is created by rasterising the sum of eggs and larvae in water tanks on the last tick.
5. **Validation**: The continuous Boyce index (Ibrahim et al., 2024) and the proportion of occurrence points falling into the top 20% of the suitability surface are computed using independently collected occurrence data (from GBIF, iNaturalist, Literature, clipped to the study area).
6. **Export**: Daily summaries and a sample of raw data are saved for further analysis.

All figures and rasters are stored in a user-defined output directory.

2.2 Case Study Design

2.2.1 Vector and Study Site

2.2.2 Vector Biology

2.2.3 Data Sources

2.2.4 Simulation Assupmtions, Rules and Configuration

3 Results

4 Discussion

5 Conclusion

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Code availability

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